

# Likelihood Function, examples

CSU, Chico Math 314

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# Bernoulli Distribution

With pen and paper or LaTeX, derive the maximum likelihood estimator for the parameter  $p$  of the Bernoulli distribution, with PDF  $f(x|p) = p^x(1-p)^{1-x}$ . Show your work.

# Weibull Distribution

With pen and paper or LaTeX, derive the maximum likelihood estimator for the scale parameter  $\lambda$  of the Weibull distribution,  $N(k, \lambda)$ , given a known shape parameter  $k = 3$ . Show your work.

$$f(x|\lambda, k = 3) = \left(\frac{k}{\lambda}\right) \left(\frac{x}{\lambda}\right)^{k-1} \exp\left(-(x/\lambda)^k\right), x \geq 0.$$

# Normal Distribution

With pen and paper or LaTeX, derive the maximum likelihood estimator for the mean  $\mu$  of the normal distribution,  $N(\mu, \sigma^2 = 9)$ . Show your work.

$$f(x|\mu, \sigma) = (2\pi\sigma^2)^{-1/2} \exp\left(\frac{-(x - \mu)^2}{2\sigma^2}\right).$$

# Normal Distribution

With R, generate data from the distribution  $N(\mu = 3, \sigma^2 = 4)$ , and then use `optim` to estimate  $\mu$  and  $\sigma^2$  via the log-likelihood function for the Normal distribution.

```
## sample from known population  
X <- rnorm(1000, mean=3, sd=2) # mu=3, sd=2  
## then assume we don't know the actual parameters  
## and estimate parameters mu and sigma
```